Daniel Xiao

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Education

University of California, Los Angeles Sep. 2025 – Jun. 2028
Bachelor of Science in Computer Science *GPA: **3.969***

- Relevant Coursework: Data Structures and Algorithms, Computer Architecture, Software Construction, Discrete Math, Linear Algebra

Technical Skills

Languages: Java, Python, C++, C, SQL, HTML/CSS, Swift
Frameworks, Libraries: React, FastAPI, PyTorch, pandas, NumPy, Matplotlib, MongoDB, Playwright, Kotlin/JS
Developer Tools: Git, AWS, Vercel, VS Code, Visual Studio, PyCharm, IntelliJ, Eclipse, Cursor, Claude Code, CUA

Experience

Software Engineer Intern June 2026 – Sept 2026
Bumblebee Networks *Mountain View, CA*

- Built a **document transfer and storage system** on **AWS** to move diagnostic logs and support ticket data between branch devices and the cloud
- Worked with **real-time WAN health monitoring** and application visibility across distributed branch deployments, helping operators catch outages before they hit the business
- Worked on **cellular branch failover** and **out-of-band (OOB) device management**, keeping branches online and remotely reachable without an on-site technician

Project Lead, Backend & AI Feb. 2026 – Aug. 2026
TreeReq *Los Angeles, CA*

- Led a **cross-functional team** of **9** engineers building a course-planning platform for UCLA students to navigate prerequisites and plan schedules
- Architected a **RESTful API** (**FastAPI** + **MongoDB**) serving prerequisite and course data to a **React** frontend
- Developed a **web scraping** pipeline with **Playwright**, **Chromium**, and the **Claude API (LLM)** to keep prerequisite data accurate across all **141** UCLA majors

Software Engineer Sept. 2025 – June 2026
UCLA IEEE *Los Angeles, CA*

- Built an **autonomous maze-solving robot** designed to navigate unknown mazes and find the exit with no human input
- Implemented **depth-first search (DFS)** and **graph traversal** with **IR sensors** and motor controllers, letting the robot sense its surroundings and path in real time
- Designed and soldered a custom **PCB** for motor control and sensor integration, iterating through multiple hardware revisions to a reliable final build

Machine Learning Research Assistant Feb. 2023 – Feb 2025
Purdue University *West Lafayette, IN*

- Developed and trained a lightweight, low-latency **deep learning model** (**5K** parameters) for **autonomous UAV control**, achieving **3.04ms** inference latency for real-time edge deployment
- Optimized obstacle-avoidance model performance through **hyperparameter tuning** across **9,920** training epochs, achieving a consistent **27cm** mean error across all test scenarios
- Co-authored a **peer-reviewed research paper** and served as **Lead Presenter** at the AIAA Aviation and Aeronautics Forum

Projects

Autonomous Rover | *Java, Arduino, CAD* Sept. 2025 – Dec. 2025

- Built an **autonomous navigation system** that pathfinds and avoids obstacles in real time without human control, using **Arduino** and Java
- Integrated gyroscope and distance sensors into a **closed-loop sensor-motor feedback** system, letting the rover adjust its path on the fly
- Designed a custom chassis in **Fusion 360 (CAD)** built around sensor placement and structural durability

MNIST Handwritten Digit Classification | *Python, PyTorch, NumPy, Matplotlib* Sept. 2025 – Dec. 2025

- Built and trained a **convolutional neural network (CNN)** achieving **98%+** classification accuracy on the MNIST benchmark dataset using **PyTorch**
- Designed and evaluated multiple **CNN architectures** (varying convolutional/pooling layer depth) to analyze tradeoffs between model accuracy, training time, and overfitting
- Visualized model performance and error patterns using **Matplotlib** and **NumPy** to guide architecture choices